

CHAPTER 3 — THEORETICAL AND CONCEPTUAL FRAMEWORK

3.1 Theoretical Basis for Behavioral Cue Analysis

Purpose: Explain the theoretical basis for using observable behavioral cues in ARGUS.

Include:

- Briefly introduce the theoretical foundations adopted by the study.
- Explain only those concepts that directly support the design of ARGUS.
- Explain why behavioral cues may provide information relevant to suspicious behavior during an examination.
- Discuss the relationship between observable behavior and interpretation.
- Clarify an important assumption of the study:
 - gaze deviation, head movement, posture changes, etc. are **behavioral cues**;
 - individually, they are not proof of cheating;
 - interpretation should consider patterns across cues and over time.
- Relate the theoretical concepts to the specific modalities used by ARGUS.

3.2 Multimodal Behavioral Representation

Purpose: Explain why ARGUS uses several behavioral modalities rather than relying on a single cue.

3.2.1 Head and Facial Orientation

Include:

- What head/facial orientation represents.
- What observable information can be obtained from it.
- Why it is relevant to examination behavior.
- Examples of potentially informative patterns:
 - repeated head turns;
 - sustained orientation away from the screen;
 - changes in head direction.
- Limitations of interpreting head orientation alone.

3.2.2 Eye Gaze

Include:

- What gaze information represents.
- Why gaze direction may be useful in examining visual attention.

- Examples of gaze behaviors considered by the system.
- Why gaze deviation alone cannot establish suspicious behavior.
- Relationship between gaze and head orientation.

3.2.3 Upper-Torso and Posture

Include:

- What information is captured from the upper body.
- Why posture and torso movement provide complementary information.
- Types of movement represented by this modality.
- Limitations of posture as an independent indicator.

3.2.4 Temporal Behavioral Patterns

Include:

- Why behavior should be examined over time rather than frame-by-frame only.
- Difference between:
 - isolated movement;
 - sustained behavior;
 - repeated behavior;
 - co-occurring behaviors.
- Why temporal aggregation is needed before multimodal inference.

3.3 Multimodal Fusion and Behavioral Inference

Purpose: Explain the conceptual rationale for combining the three modalities.

Include:

- Definition of multimodal fusion in the context of ARGUS.
- Why one modality may be ambiguous.
- How complementary modalities can provide additional context.
- Conceptual relationship among:
 - gaze;
 - head/facial orientation;
 - posture/upper-torso movement.
- Explain how multiple cues occurring together may provide stronger behavioral evidence than one cue alone.
- Introduce the role of the **Multimodal Behavior Inference Engine (MBIE)**.
- Explain that MBIE converts extracted observations into a structured multimodal behavioral representation for subsequent analysis.

Do **not** put detailed equations, thresholds, implementation parameters or code here. Those belong in Chapter 4.

3.4 ARGUS Conceptual Framework

Purpose: Present the complete conceptual flow of the proposed system.

Include a revised conceptual framework/system-level diagram showing:

Recorded Examination Video

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Video Preprocessing and Segmentation

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Parallel Feature Extraction

- Head/Facial Orientation
- Eye Gaze
- Upper-Torso/Posture

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Temporal Alignment and Multimodal Fusion

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Multimodal Behavior Inference Engine (MBIE)

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Structured Behavioral/Risk Representation

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LLM-Based Behavioral Analysis

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Behavioral Risk Report

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Human Review

For every block:

- explain its purpose;
- identify its input;
- identify its output;

- explain how its output becomes the input of the next component.

The framework should represent the **system that was actually implemented**, not the system originally envisioned in the proposal.

3.5 Behavioral Risk Report and Decision Support

Purpose: Explain the conceptual role of the final ARGUS output.

Include:

- What the Behavioral Risk Report contains.
- How detected behavioral events are represented.
- Role of timestamps.
- Role of LLM-generated interpretation/reasoning.
- Meaning of the suspicious/non-suspicious assessment.
- How the report can assist a human reviewer.

Include a sample report or simplified example if useful.

3.6 Role and Boundaries of ARGUS

Purpose: Clearly define what the system claims—and does not claim—to do.

Include:

- ARGUS is a **proof-of-concept behavioral-suspicion classification system**.
- It analyzes observable behavioral cues.
- It does not establish whether academic dishonesty actually occurred.
- A flagged behavior indicates that a segment/session may warrant human review.
- Behavioral cues may also occur naturally during legitimate examination behavior.
- ARGUS is intended as decision support rather than an autonomous adjudication system.

This is consistent with the distinction the thesis already makes between behavioral suspicion and confirmed cheating.

CHAPTER 4 — METHODOLOGY

Chapter 4 should answer:

What exactly did you do, and how could another researcher reproduce your work?

Everything here should describe the final implementation and actual experiment, not what was originally planned.

4.1 Research Design

Include:

- Type of study.
- Overall experimental design.
- Controlled/simulated examination setting.
- Offline proof-of-concept evaluation.
- Overall research workflow.
- Experimental units:
 - participant;
 - recording/session;
 - segment.
- Explain which unit is used for which analysis.
- Define the two scripted conditions:
 - cheating/suspicious condition;
 - non-cheating/benign condition.
- Explain the distinction between a **scripted condition label** and an independently verified behavioral event.
- Briefly introduce the experiments conducted.

4.2 Dataset Construction and Experimental Setup

4.2.1 Participants and Recording Setup

Report what **actually occurred**.

Include:

- Number of participants.
- Number of unique participants.
- Number of sessions.
- Relationship between participants and sessions.
- Inclusion criteria.

- Recording location/environment.
- Webcam specifications.
- Resolution.
- Frame rate.
- Camera placement.
- Lighting/background conditions.
- Approximate recording duration.
- Examination setup.
- Any deviations from the planned procedure.

Do not retain numbers from the proposal if the actual study used different numbers.

The completed study reports 11 sessions and 154 processing segments, rather than simply the originally planned sample.

4.2.2 Mock Examination Scenarios

Include:

- How the mock examination was conducted.
- How scenarios were assigned.
- Description of cheating/suspicious scenarios.
- Description of non-cheating/benign scenarios.
- Expected behaviors associated with each scenario.
- Rationale for including natural behaviors that could potentially cause false positives.

Provide a table:

Scenario	Condition	Expected/Scripted Behavior	Behavioral Modality Targeted
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4.2.3 Reference Labels and Ground Truth

Include:

- How the reference labels were established.
- Who assigned the labels.
- Whether labels were predetermined by scenario.
- Whether labels were assigned at:
 - participant level;
 - session level;
 - segment level;
 - event level.
- Explain how segment labels were derived from session labels.
- Explain exactly what was manually annotated.
- Explain what ELAN was used for.
- Explicitly state what **was not independently annotated**.

The current document already states that the EyeGaze ELAN tier was used for limited qualitative checking rather than comprehensive multimodal ground-truth annotation.

Do not call the labels "ground truth behavioral events" if they merely represent assigned experimental conditions.

4.2.4 Video Segmentation

Include:

- Why recordings were segmented.
- Segment duration.
- Treatment of final partial segments.
- How timestamps were retained.
- Number of segments generated.
- Whether each segment was processed independently.
- Relationship between segment-level observations and session-level classification.

4.3 ARGUS System Architecture and Processing Pipeline

Purpose: Describe the actual implemented system.

Include:

- System architecture diagram.
- Technologies/libraries used.
- Overall data flow.
- Input/output format for each major component.

4.3.1 Video Input and Preprocessing

Include:

- Video loading.
- Frame extraction/sampling.
- Resolution handling.
- Frame rate handling.
- Landmark detection requirements.
- Handling of missing/failed detections.
- Any normalization performed.

4.3.2 Head and Facial Orientation Feature Extraction

For **every feature**, explain:

source landmarks → **computation** → **threshold/rule** → **resulting feature** → **behavioral interpretation**

Include:

- model/library;
- landmarks used;
- measurements calculated;
- direction categories;
- thresholds;
- temporal aggregation;
- final output variables.

Include equations where applicable.

4.3.3 Eye-Gaze Feature Extraction

Again describe:

landmarks → **computation** → **classification** → **temporal aggregation** → **output**

Include:

- eye landmarks;
- gaze calculation;
- direction categories;
- thresholds;
- handling of unavailable gaze estimates;
- segment-level aggregation;
- output features.

4.3.4 Upper-Torso/Posture Feature Extraction

Include:

- pose-estimation model.
- body landmarks used.
- geometric measurements.
- torso/shoulder relationships.
- movement measurements.
- thresholds.

- temporal aggregation.
- output features.

4.4 Multimodal Behavior Inference Engine (MBIE)

4.4.1 Temporal Alignment

Include:

- How outputs from the three pipelines are synchronized.
- Common time unit.
- Handling of missing modality values.
- Relationship between frame-, second-, and segment-level information.

4.4.2 Feature Aggregation

Include:

- Which features enter MBIE.
- How they are summarized.
- Any counts, proportions, durations, frequencies, maxima, means, etc.
- Why each selected feature is retained.

4.4.3 Rule-Based Risk Processing

Include:

- All implemented rules.
- Thresholds.
- Weights, if any.
- Conditions for generating behavioral events.
- How multiple modalities are combined.
- How risk-related values are derived.

For **every threshold/rule**, identify whether it came from:

- literature;
- empirical calibration;
- researcher-defined heuristic.

Explain the rationale.

4.4.4 Construction of the Multimodal Risk Representation

Include:

- Structure of the resulting representation/JSON.
- Meaning of every important field.
- How the LLM uses it.

Provide **one complete worked example**:

video segment → extracted features → MBIE processing → final structured representation.

4.5 LLM-Based Behavioral Analysis

4.5.1 LLM Input Representation

Include:

- Exactly what information is sent to the LLM.
- What information is *not* sent.
- Whether raw video/images are supplied.
- Structure of the prompt input.
- How timestamps and modality evidence are represented.
- Confirm that the reference label is withheld from the LLM during inference.

4.5.2 ARGUS-Lite Prompt

Include:

- Purpose of the prompt.
- Prompt design principles.
- Required output format.
- Instructions given to the LLM.
- Version used for the final experiment.
- Why a fixed prompt was used.

Put the **complete prompt** in the appendix.

4.5.3 LLM Models

For each model:

- exact model name/version;
- access method;
- relevant inference settings, if known;
- date/version of evaluation;
- whether identical input and prompt were used.

4.5.4 Behavioral Risk Report Generation

Include:

- Output fields.
- Classification.
- Evidence.
- Timestamped events.
- Reasoning/interpretation.
- Evidence-strength/risk descriptions.
- How the final report is generated and stored.

4.6 Experimental Evaluation

4.6.1 Experiment 1 — Verification of Feature Extraction

Research question:

Do the extracted features correspond to observable behavior in the recorded videos?

Include:

- Manual verification procedure.
- Use of ELAN.
- Number and method of selecting timestamps/segments.
- Features/modalities checked.
- Criteria for agreement/disagreement.

If verification was limited, explicitly call it **qualitative spot-checking**, not comprehensive validation.

4.6.2 Experiment 2 — Conventional Classification Baseline

Include:

- Purpose of the baseline.
- Logistic regression configuration.
- Input features.
- Why those features were selected.
- Missing-value treatment.
- Imputation.
- Standardization.
- Regularization.
- Class weighting.

- Cross-validation procedure.
- Unit held out during each fold.
- Measures taken to avoid data leakage.

4.6.3 Experiment 3 — LLM-Based Classification

Include:

- Qwen experiment.
- OpenAI o3 experiment.
- Fixed prompt.
- Input representation.
- Number of sessions processed.
- How the final prediction was obtained from each report.
- How predictions were compared against reference labels.

4.6.4 Evaluation Metrics

Define and justify:

- True Positive
- True Negative
- False Positive
- False Negative
- Accuracy
- Precision
- Recall/Sensitivity
- Specificity
- F1-score
- Confusion matrix
- ROC-AUC, **only for models producing an appropriate continuous score**

Include equations.

4.6.5 Comparative Analysis

Explain how the following are compared:

- logistic regression baseline;
- Qwen3.7-Plus;
- OpenAI o3.

Clarify that this is a comparison **within this dataset and experimental setting**, not evidence that one approach is generally superior.

4.7 Ethical Considerations

Include:

- informed consent;
- video privacy;
- storage/access;
- anonymization;
- simulated rather than actual cheating;
- handling of participant identities;
- risks of false positives;
- risks of automated behavioral interpretation;
- importance of human review;
- limitations on using the prototype for disciplinary decisions.

CHAPTER 5 — RESULTS AND ANALYSIS

Do not merely show tables and figures. Every important result must be followed by an interpretation: What happened? Why might it have happened? What evidence supports that explanation? What does it mean for ARGUS?

Do not introduce a new experimental procedure in Chapter 5. If you performed an analysis, its procedure must first be described in Chapter 4.

5.1 Dataset and Experimental Outcome

Report:

- number of unique participants;
- number of sessions;
- sessions per condition;
- video durations;
- number of segments;
- segments per session;
- missing/unusable data;
- feature-extraction failures, if any.

Provide a concise dataset summary table.

Do not interpret classification performance yet.

5.2 Feature Extraction Results

5.2.1 Head and Facial Orientation Results

Present:

- resulting features;
- descriptive statistics;
- distributions;
- representative examples;
- differences observed between conditions.

Then discuss:

- Which behaviors were captured well?
- Which were difficult?
- Were there unexpected patterns?

5.2.2 Eye-Gaze Results

Same structure:

results → **observations** → **interpretation**

5.2.3 Upper-Torso/Posture Results

Same structure.

5.2.4 Cross-Modality Observations

Examine:

- co-occurring behavioral cues;
- whether modalities provide complementary information;
- examples where modalities agree;
- examples where modalities disagree;
- exploratory differences between scripted conditions.

Clearly distinguish **exploratory patterns** from statistically/generalizably established effects.

5.3 Verification of Extracted Behavioral Cues

Present the manual/ELAN checks.

Recommended table:

Session/Timestamp	Human Observation	ARGUS Output	Agreement	Interpretation
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Include examples of:

- correct detection;
- missed detection;
- ambiguous detection;
- false behavioral trigger, if available.

Discuss:

- what appears reliable;
- where feature extraction fails;
- likely technical causes;
- limitations of the verification procedure.

The existing thesis contains an illustrative EXAM-P01 check around 02:07, but correctly identifies it as a spot-check rather than complete multimodal validation.

5.4 MBIE and Multimodal Behavioral Representation Results

Show what happens **between feature extraction and the LLM**.

Include representative cases:

Case A — Scripted Suspicious Session

Show:

modality features → MBIE → risk representation

Case B — Scripted Non-Suspicious Session

Same.

Case C — Difficult/Ambiguous Session

If available.

Discuss:

- which features contribute to the resulting representation;
- how modalities interact;
- whether MBIE preserves meaningful behavioral information;
- cases where its rules may amplify benign behaviors.

5.5 Conventional Classifier Baseline Results

Present:

- session-level predictions;
- confusion matrix;
- performance metrics;
- relevant classifier scores.

Then analyze:

- correctly classified sessions;

- false positives;
- false negatives;
- which feature patterns characterize errors;
- whether certain modalities/features appear particularly influential.

Do not claim causality simply from association.

5.6 LLM-Based Behavioral Analysis Results

5.6.1 Qwen3.7-Plus Results

Present:

- predictions;
- confusion matrix;
- metrics;
- representative Behavioral Risk Reports.

Analyze:

- correct cases;
- false positives;
- false negatives;
- evidence used by the LLM.

5.6.2 OpenAI o3 Results

Use the same structure.

5.6.3 Comparison of the Two LLMs

Compare:

- classifications;
- types of errors;
- sensitivity;
- specificity;
- evidence interpretation;
- cases where they agree;
- cases where they disagree.

Do not merely say which has the larger metric. Explain the behavioral pattern of the errors.

5.7 Error Analysis

This should be one of the major analytical sections of the thesis.

Organize cases as:

- True Positives
- True Negatives
- False Positives
- False Negatives

For important cases, trace the decision backward:

Final classification

↑

LLM interpretation

↑

MBIE risk representation

↑

Extracted multimodal features

↑

Observed behavior in original video

Ask for every error:

- What did the participant actually do?
- What did ARGUS extract?
- Which thresholds/rules were triggered?
- What did MBIE communicate to the LLM?
- How did the LLM interpret that information?
- Where in the pipeline did the error likely originate?

Separate:

feature-extraction error → fusion/rule error → LLM interpretation error

whenever the evidence allows them to do so.

5.8 Comparison of LLM Reasoning and the Conventional Baseline

Present the three approaches together:

Measure	Logistic Regression	Qwen	o3
Accuracy			
Precision			
Recall			
Specificity			
F1			

The current final document reports that the conventional baseline correctly classified 9/11 sessions while each LLM agreed with 5/11 reference labels, with particularly poor specificity for the LLMs.

Discuss:

- Why might the simpler baseline perform differently?
- What kinds of errors dominate the LLM outputs?
- Does the LLM appear overly sensitive to risk cues?
- Are benign behaviors being interpreted as suspicious?
- Is useful information lost or distorted between feature extraction and LLM input?
- What evidence in the actual cases supports these explanations?

Do not force an explanation if the available evidence cannot establish it. Label plausible but untested explanations as such.

5.9 Discussion According to the Research Objectives

This section should explicitly return to the four objectives.

5.9.1 Objective 1 — Dataset Construction and Annotation

Discuss:

- What was successfully collected?

- What labeling/annotation was actually completed?
- What was originally intended but not achieved?
- How does limited annotation affect interpretation?

5.9.2 Objective 2 — Multimodal Feature Extraction

Discuss:

- Were the three modalities successfully implemented?
- What evidence demonstrates that they work?
- Which aspects appear reliable?
- What limitations were observed?

5.9.3 Objective 3 — Multimodal Fusion and LLM-Based Inference

Separate two questions:

Was the pipeline successfully implemented?

from

Did the pipeline classify sessions reliably?

These are not the same question.

Discuss MBIE, structured representation, successful LLM report generation, and observed classification behavior.

5.9.4 Objective 4 — System Evaluation and Feasibility

Discuss:

- overall performance;
- false-positive behavior;
- comparison with baseline;
- strengths;
- weaknesses;
- what the experiment establishes;
- what it does **not** establish.

5.10 Overall Findings and Chapter Summary

End Chapter 5 with a short synthesis—not another repetition of every metric.

Answer:

What did this study ultimately learn from building and evaluating ARGUS?

For example, their evidence may support the conclusion that:

- multimodal behavioral cues can be extracted and represented;
- the complete ARGUS pipeline can operate from recorded video through report generation;
- the current LLM reasoning stage does not provide reliable behavioral-suspicion classification;
- false positives are a significant issue;
- the small simulated dataset and limited behavioral annotation constrain conclusions.